

CMPT 276 - Phase 2 Report (Milestone 1)

Project Group 14

Joshua Kim, Tim Yeh, Rodrigo Anasco, Sarah Herb

1. Overview of our Game (see statement for more in depth overview)

Our project is a 2D arcade-style game in which the player controls Doug, a hungry dog navigating a maze to collect food while avoiding enemies and punishments. The game consists of various maps, each with distinct rewards and punishments, depending on the map's theme (e.g., city, grassland). The objective is to collect all treats and reach the exit without losing to enemies or consuming punishments.

2. Approach to tackle milestone 1 of phase 2

Initial Planning: We begin by meeting as a team to discuss phase 2 of our project. Through this meeting, we've developed a better understanding of what this phase is asking us to do, and how we will approach what lies ahead. Similarly to assignment 2, we plan to use the Factory Method design pattern as this will make it easier to create game objects like enemies, rewards, mazes, etc.

Class Design from UML: We will create the main classes based on our UML diagram which are: Doug, Enemy, Reward, and Punishment. Each of these classes handle the responsibilities of our game such as Doug's movements, enemies movements, collecting food, etc. This will be a good starting point for our project and as we progress, we will likely change things around if needed, add various methods, classes, etc.

Management Process/Division of roles: We divide the roles according to classes as below:

- 1. Doug: playable character, reward counter - Josh
- 2. Game mechanics, some UI - Sarah
- 3. Enemies: different types of enemies, obstacles - Tim
- 4. System: Obstacles and punishments - Rodrigo

We will work on the classes related to our division of work above, and any classes that are linked between one another will require group mates to communicate about them, and work together.

Goals for Milestone 1: We hope to complete a few things for the mid-way deadline:

- Finish some of this report, especially the part of the report focused on milestone 1. We'll likely leave a good chunk of our report for milestone 2, regarding overall changes, modifications, issues faced, etc.
- Implement the classes from our UML: We will attempt to get as much of the implementation done in milestone 1, so that we can build and fix more in milestone 2. We are hoping aspects of our project such as the UI, map, exit points, game mechanics, and overall main game loop will be more focused on milestone 2. As we may be busy now and then, if ultimately needed we can have some leeway for milestone 2 if certain things could not get done (but this will unlikely be the case).

Adjustments/Modifications: Created a GameObject class: All game objects are a subclass of GameObject. We took some inspiration from a source (listed below in sources) and based some of our project around it. Thus, we may have strayed away from some of what we originally had in our UML, but was needed.

Measures to enhance quality/Challenges faced: We modularized with hopes of making our project and code more organized, easier to understand, test and maintain. In terms of a challenge, perhaps some slight miscommunication in what was getting done, as some parts of our work may have overlapped, thus required us to change around what we originally worked on to fit the project.

Sources for milestone 1:

Backend/game mechanics based on:

<https://www.youtube.com/watch?v=1gir2R7G9ws>